

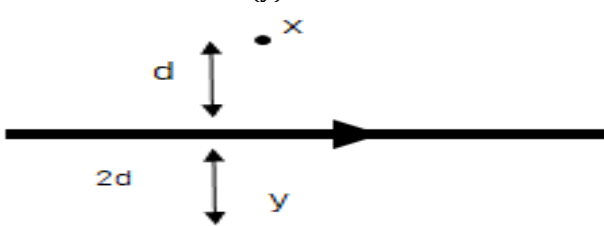
Answer the following groups

Group 1 :

A) Write the scientific term expressed by the following statements:

	Statement
1	It is the magnetic flux density which exerts a force of 1 N on a wire of 1m length carrying a current of intensity 1A placed perpendicular to the magnetic field
2	If the thumb points in the direction of the current, the rest of the fingers around the wire will indicate the direction of the magnetic field due to the current.

B) Put a mark (✓) if the statement is correct and an (X) mark if the statement is incorrect

	Statement
3	When direction of the current flow reversed, the direction of magnetic field will be reversed (.....)
4	In the following figure The ratio between the magnetic flux density at point (x) to that at point (y) $\frac{B(x)}{B(y)} = \frac{1}{2}$  (.....)
5	The magnetic flux density increases at the center of a circular coil carrying an electric current by decreasing cross sectional area of the coil (.....)

C) Put a circle around the correct choice in the following:

	Statement
6	When an electric current passes through a solenoid, the flux lines at the axis of the solenoid are..... A) Uniform concentric circles. B) Curved lines perpendicular to the coil plane. C) Straight lines parallel to the coil axis. D) Straight lines perpendicular to the coil axis.
7	The magnetic flux density at a point located vertically at a distance d from a straight wire carrying a current of intensity I equals B . When increasing the current intensity to double, then The magnetic flux density becomes... A) $0.5B$. B) $2B$. C) B . D) $4B$.
8	A straight wire 30 cm long through which passes a current 10 A. The wire is placed perpendicular to a magnetic field of magnetic flux density 0.5 T. Then the force acting on the wire (a) 1.5 N (b) 150 N (c) 0.15 N (d) 1500 N

D) Compare between magnetic torque and magnetic dipole moment (with respect to factors affecting on each of them) (two factor only)

	comparison	
	Magnetic torque	Magnetic dipole moment
9		
10		

Group 2 :

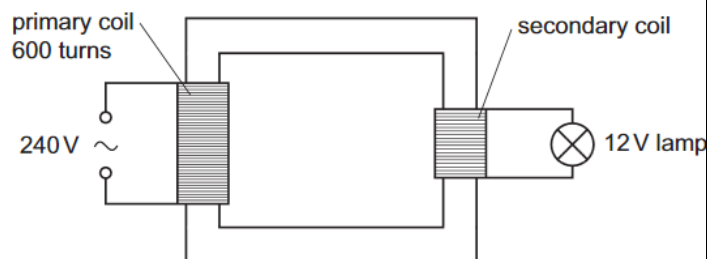
A) Put circle around the correct answer

11	A coil of copper wire is moved parallel to a uniform magnetic field. The e.m.f. induced in the coil will: a) depend on the intensity of magnetic field b) depend on the velocity of the coil c) be zero d) be infinite
12	Henry is the unit of the a) Coefficient of mutual inductance only b) Coefficient of self-inductance only c) Both a and b d) None

13 A step-down transformer is used to light a 12 V lamp from a 240 V mains supply. The lamp lights at normal brightness. The primary coil has 600 turns. How many turns are in the secondary coil?

- a) 12
c) 30

- b) 20
d) 50



B) Write the scientific term expressed by the following statements:

	Statement
14	Device used convert mechanical energy into electrical energy
15	They are induced currents that circulate in closed paths due to the change in magnetic flux through a solid conductor associating with heating effect.

C) Correct the underlined word:

	Statement
16	The idea of <u>electric motor</u> is based on the principle of electromagnetic induction,
17	If the number of turns of the metal coil cutting the variable magnetic flux increases at a constant rate, then the average induced electromotive for <u>decreases</u>

D) Put a mark (✓) if the statement is correct and an (X) mark if the statement is incorrect

	Statement
18	When the rate of change in the current intensity in a coil increases into double, the self-inductance coefficient of the coil doubled. (.....)
19	Switching the current in the primary coil on and off will produce an induced electromotive force in the secondary coil. (.....)
20	A straight wire 1 m long is moving in a uniform magnetic field of flux density 0.1 T at a uniform velocity 10 m/s. an induced potential difference of 0.5 V is generated between its terminals. Then the angle between the direction of the wire motion and the direction of the magnetic field is. 30°. (.....)

Group 3 :**A) Write the scientific term expressed by the following statements:**

	Statement
21	The wavelength at maximum radiation intensity (λ_{max}) is inversely proportional to the temperature
22	The ratio between the photon energy to its frequency

B) Correct the underlined word:

	Statement
23	Wavelength λ of visible light ranges from <u>100</u> nm to 700 nm.,
24	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the photon's speed <u>decreases</u>
25	The process in which electrons are emitted from a metallic surface when a light given to it is called <u>thermionic effect</u>

C) Put circle around the correct answer

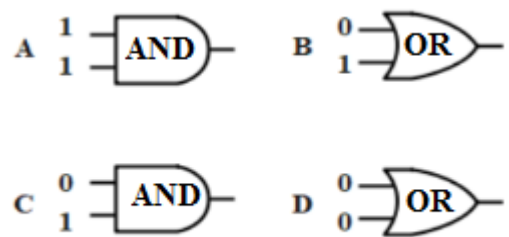
26	The Characteristic radiation of X rays is depend on ... a) The kind of target element. b.)The voltage difference between the filament and the target c.) a and b are correct d) None of these
27	From the characteristic of laser beam a) Monochromatic b)non coherence c) varying in phase d. all of the above
28	population inversion state that the number atoms in the excited energy state is the number of atoms lower energy state a) less than b) greater than c) equal d) none

D) Compare between kinetic Energy of electron and its momentum with respect to mathematical relation:

	Statement	
29	kinetic Energy of electron	momentum of photon
30		

Group 4 :

A) Put circle around the correct answer

31	When connecting the diode in forward direction, a) The width of depletion layer increases and diode resistance decreases. b) The width of depletion layer decreases and diode resistance decreases. c) Neither the potential barrier width of depletion layer nor the diode resistance changes. d) The width of depletion layer increases and the diode resistance increases.
32	When the temperature of pure silicon crystal (Si) lowered, so its electrical conductivity..... a) decreases b) does not change c) increases d) vanishes
33	Which of the following gates its output is 0? a) Only B b) Only D c) Only A d) C and D 

B) Compare between each of the following

34 35	Point of comparison	NOR gate			NAND gate		
	Complete the truth table	Input A	Input B	Output	Input A	Input B	Output
		1	0		1	1	

C) Correct the underlined word:

36	P-type semiconductor is a semiconductor which is obtained by doping with <u>pentavalent</u> impurity atoms
37	In NPN transistor the emitter current is the <u>multiplication</u> of collector current and base current
38	Crystal of pure Silicon consists of Silicon atoms bound together in <u>ionic</u> bond

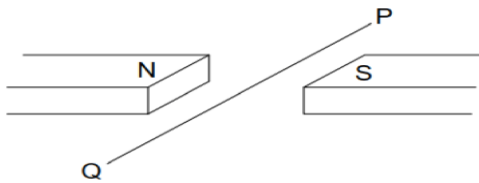
D) Write the scientific term expressed by the following statements:

	Statement
39	The number of bonds broken per second equal to the number of bonds mended per second
40	The circuits that perform logic operations and depend on digital electronics

Group 5 :

Answer the following

- 41 A length of 0.25 m of the wire is subject to a magnetic field strength of 0.04 T. A
42 downward magnetic force of 0.06 N acts on the wire. **What is the magnitude of the current?**



- 43 A certain solenoid consists of 100 turns of wire and has a length of 0.1m. **Find the**
44 **magnitude of the magnetic flux density** inside the solenoid when it carries a current of 0.5A. (given that magnetic Permeability of air ($\mu_{\text{air}} = 4\pi \times 10^{-7} \text{ Wb/m.A}$)

- 45 A rod of copper of length 0.2 m moves with velocity 0.5m/second in a
46 perpendicular direction to a magnetic field of density 0.8T. Calculate the emf induced in such a rod

- 47 A black body is heated to a temperature 5800° K so the wavelength that is
48 associating with its radiation of the maximum intensity was 500 nm ,
Calculate the wavelength of maximum intensity if the temperature becomes 6000° K

- 49 Find The current gain (β) of a transistor in common
50 emitter mode as shown in the figure. **using data on figure**

